📘 **AI/ML Internship – Day 39 Notes & Tasks**

🟢 **Module 7: K-Nearest Neighbors (KNN) Classification Algorithm**

📌 **Part 1: Introduction to KNN**

🔹 **What is KNN?**

👉 K-Nearest Neighbors (KNN) is a Supervised Machine Learning algorithm used for:

- Classification
- Regression

👉 Most commonly used for Classification problems.

🔹 **Full Form**

KNN = K-Nearest Neighbors

📌 **Part 2: Real-Life Example**

Imagine you move to a new city.

You want to know if a restaurant is good.

Instead of checking the restaurant yourself, you ask nearby people.

If most nearby people say:

✓ Good → You assume it is good.

✓ Bad → You assume it is bad.

👉 KNN works in the same way.

It looks at the nearest data points and decides based on majority voting.

📌 **Part 3: Theory Answers (IMPORTANT)**

🔹 **1. What is KNN?**

👉 KNN is a supervised machine learning algorithm that classifies data based on nearest neighbors.

🔹 2. What does "K" mean?

👉 K represents the number of nearest neighbors considered.

Example:

K = 3

The algorithm checks the 3 nearest data points.

🔹 3. Is KNN Supervised or Unsupervised?

👉 KNN is a Supervised Learning algorithm.

🔹 4. What is a Neighbor?

👉 A neighbor is a nearby data point used to make predictions.

🔹 5. Why is KNN Popular?

👉 Because:

- Easy to understand
- Easy to implement
- Good for small datasets

📌 Part 4: How KNN Works

Step 1

Choose K value.

Example:

K = 3

Step 2

Find nearest data points.

Step 3

Count categories.

Step 4

Select majority class.

📌 Part 5: Example Dataset

Study Hours	Result
1	Fail
2	Fail
3	Fail
5	Pass
6	Pass
7	Pass

Suppose:

New Student = 4 Hours

K = 3

Nearest Students:

3 Hours → Fail

5 Hours → Pass

6 Hours → Pass

Result:

Pass = 2

Fail = 1

Prediction:

👉 PASS

📌 Part 6: Visual Representation

Study Hours

1(F)   2(F)   3(F)   4(?)   5(P)   6(P)   7(P)

Nearest Neighbors:

3(F)
5(P)
6(P)

Majority = PASS

📌 Part 7: Install Scikit-Learn

pip install scikit-learn

📌 Part 8: Import Libraries

import pandas as pd

from sklearn.neighbors import KNeighborsClassifier

📌 Part 9: Create Dataset

```
data = {
  "Hours":[1,2,3,5,6,7],
  "Result":[0,0,0,1,1,1]
}

df = pd.DataFrame(data)
```

```
print(df)
```

Meaning

Value	Meaning
0	Fail
1	Pass

📌 Part 10: Features and Labels

```
X = df[["Hours"]]

y = df["Result"]
```

📌 Part 11: Create KNN Model

```
model = KNeighborsClassifier(n_neighbors=3)
```

Explanation

n_neighbors = 3

Means:

Check 3 nearest neighbors.

📌 Part 12: Train Model

```
model.fit(X, y)
```

📌 Part 13: Make Prediction

```
prediction = model.predict([[4]])

print(prediction)
```

Output:

[1]

Meaning:

👉 PASS

📌 Part 14: Predict Multiple Values

```
prediction = model.predict([[2],[4],[8]])

print(prediction)
```

Possible Output:

[0 1 1]

Meaning:

Hours	Prediction
2	Fail
4	Pass
8	Pass

📌 Part 15: Complete Program

```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier

data = {
   "Hours":[1,2,3,5,6,7],
   "Result":[0,0,0,1,1,1]
}

df = pd.DataFrame(data)

X = df[["Hours"]]
y = df["Result"]
```

```
model = KNeighborsClassifier(n_neighbors=3)

model.fit(X, y)

prediction = model.predict([[4]])

print("Prediction:", prediction)
```

📌 Part 16: Choosing K Value

K Value	Effect
Small K	Sensitive to noise
Large K	More stable
Very Large K	May reduce accuracy

Common values:

K = 3
K = 5
K = 7

📌 Part 17: Advantages of KNN

✅ Simple to understand

✅ Easy implementation

✅ No complex training

✅ Good for small datasets

📌 Part 18: Disadvantages of KNN

❌ Slow for large datasets

❌ Requires more memory

❌ Sensitive to irrelevant features

📌 Part 19: Real-World Applications

Education

👉 Pass/Fail Prediction

Healthcare

👉 Disease Classification

Banking

👉 Loan Approval Prediction

Security

👉 Face Recognition

E-Commerce

👉 Customer Classification

📌 Part 20: Difference Between Logistic Regression and KNN

Logistic Regression	KNN
Learns mathematical relationship	Uses nearest neighbors
Fast prediction	Slower prediction
Model-based	Instance-based
Works well on large datasets	Better for small datasets

🎯 Tasks for Day 39

🧠 Task 1: Theory

1. What is KNN?

2. What does K mean in KNN?
3. Is KNN supervised or unsupervised?
4. What is a neighbor?
5. Advantages of KNN?

💻 Task 2: Dataset Practice

1. Create student result dataset
2. Create disease dataset
3. Create loan approval dataset

💻 Task 3: KNN Model

1. Train KNN model
2. Use K = 3
3. Predict results

💻 Task 4: Experiment with K Values

1. K = 1
2. K = 3
3. K = 5

Compare outputs.

💻 Task 5: Classification Practice

Predict:

1. Student result
2. Disease status
3. Customer category

💻 Task 6: Comparison Activity

Compare:

1. Logistic Regression
2. KNN

Write differences.